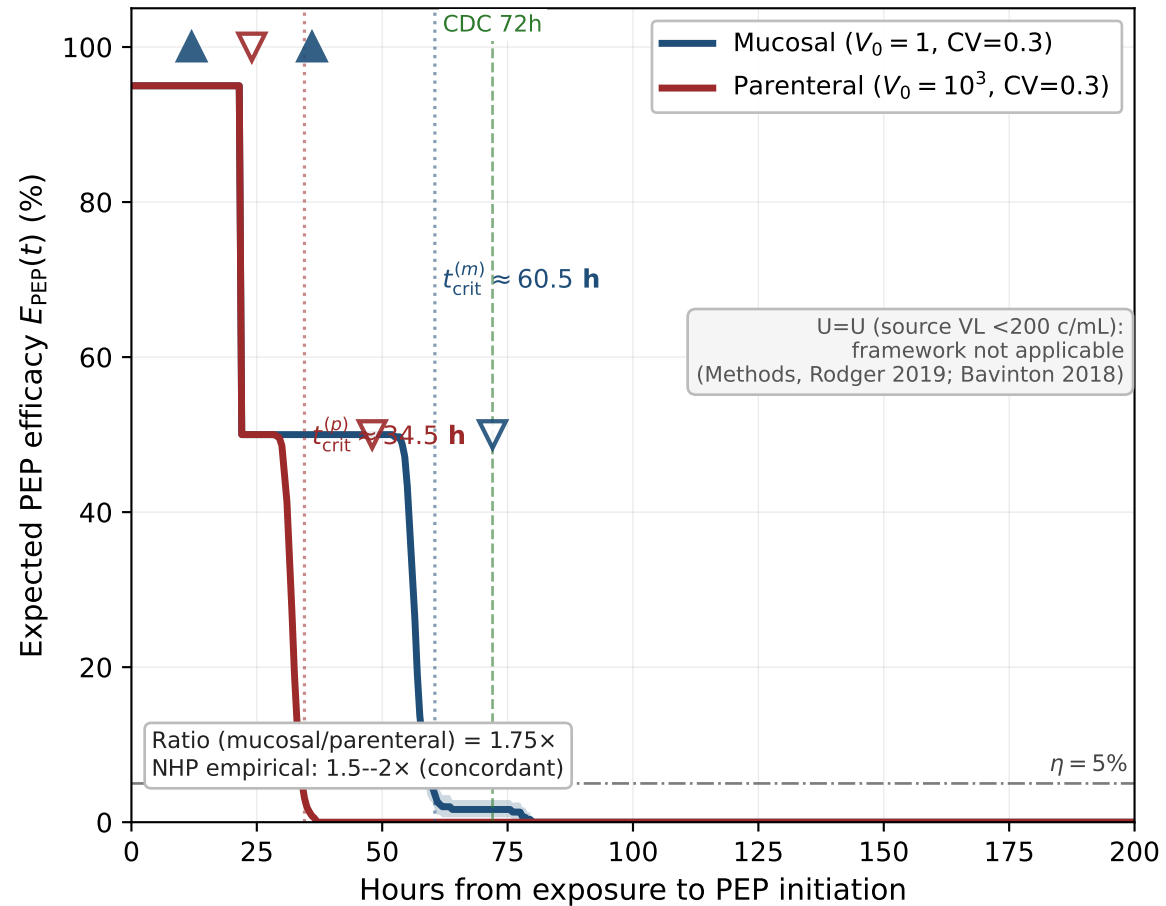
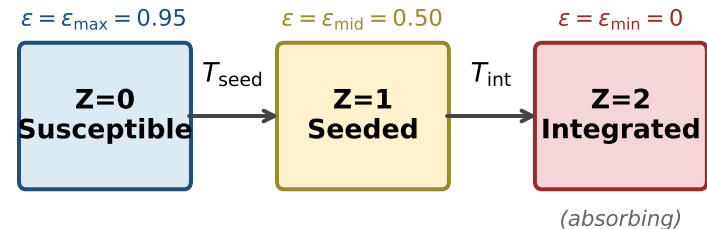


A. Route comparison: emergent t_{crit} from multiscale dynamics



B. Three-state within-host model & route-dependent inoculum



Route-dependent inoculum



By inoculum-monotone hitting times (Lemma S5.2): $T_{\text{int}}^{(p)} \leq T_{\text{int}}^{(m)}$

$$\Rightarrow t_{\text{crit}}^{(p)} \leq t_{\text{crit}}^{(m)} \text{ (Corollary S5.1)}$$